

Executive Summary

Customer Segmentation & Churn Analytics using Machine Learning

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Customer Segmentation & Churn Pattern Analytics in European Banking

This project analyses customer churn patterns in a European banking dataset containing 10,000 customer records across France, Germany, and Spain. The primary objective was to identify customer segments associated with different levels of observed churn and examine patterns across geography, age, tenure, credit score, balance, gender, customer engagement, and product usage.

The analysis was conducted using Python, Pandas, Matplotlib, and Seaborn, with customer segmentation created according to the project requirements. Additional analysis focused on high-balance customers and customer engagement. An interactive Streamlit dashboard was developed to allow users to explore the results through filters and dynamic KPIs.

The overall observed churn rate was 20.37%, with 2,037 customers classified as churned. Geographic analysis showed an observed churn rate of 32.44% in Germany, compared with 16.15% in France and 16.67% in Spain. Age-based analysis identified the 46–60 age segment as having the highest observed churn rate at 51.12%. Customers classified as high-balance customers had an observed churn rate of 24.98%, indicating greater customer exposure within this segment.

The analysis also identified an Engagement Drop Indicator of 12.58 percentage points, based on the difference between observed churn rates for inactive and active customers. New customers showed an observed churn rate of 21.14%, while customers in the low credit-score category showed an observed churn rate of 21.75%.

Based on these patterns, the analysis highlights areas that can be examined for customer retention efforts, including country-specific customer experience, engagement among older customer segments, monitoring of higher-balance customers, early-stage customer engagement, and further investigation of lower credit-score segments. These observations can support more targeted customer analysis rather than relying only on overall churn figures.

The project delivers a research paper, analytical visualizations, an interactive Streamlit dashboard, and supporting analytical files. The dashboard provides dynamic KPIs, segmentation filters, high-value customer analysis, engagement analysis, financial analysis, and filtered-data download functionality.

Overall, the project demonstrates how customer segmentation and exploratory analytics can be used to identify where churn is concentrated and which customer characteristics are associated with higher observed churn. The findings represent patterns within the available dataset and do not establish causal relationships. Further analysis using transaction history, customer behaviour, product-level information, and longitudinal data could provide deeper insight into the underlying drivers of churn.

Conclusion:

The analysis demonstrates that customer churn is not evenly distributed across the European banking customer base. The observed differences across geography, age, balance, tenure, credit score, and engagement provide useful indicators for identifying customer segments that require closer examination. The Streamlit dashboard further converts these findings into an interactive format, allowing users to explore segments and key performance indicators dynamically.

The project provides a structured analytical framework for understanding churn patterns and supporting data-driven customer retention discussions. However, the findings represent observed associations within the available dataset and do not establish causal relationships. Additional information such as transaction history, customer interactions, product usage, and longitudinal behaviour could provide deeper understanding of the factors associated with customer churn.